

FALL 2026



EE3003 Analog and Digital Communications

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Department of Electrical Engineering

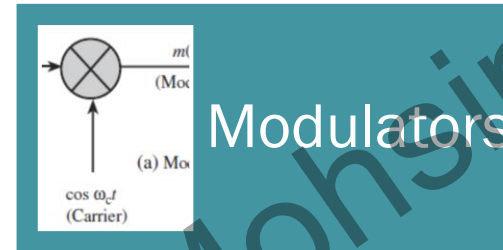
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Lecture Objectives

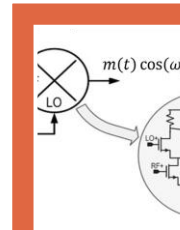
- Demodulation of DSB-SC Signals
- Amplitude Modulation with Carrier
- Practical Circuits for AM generation
- Demodulation of AM (Coherent and Non-Coherent)
 - Rectifier Detector
 - Envelope Detector

Demodulation of DSB-SC Signals

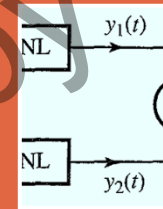
- In demodulator, **DSB-SC signal is the input** and the multiplier output is passed through a **LPF**.



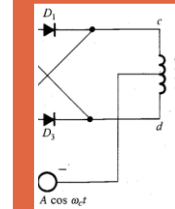
- In modulator, **$m(t)$ is the input signal** and the multiplier output is passed through a **BPF** tuned to ω_c .



1. Electronic Multipliers



2. Non-linear Modulators



3. Switching Modulators

- Demodulation of DSB-SC signals essentially involves multiplication with the carrier signal and is identical to modulation.
- The only difference between them lies in the i) input signal and ii) the output filter.
- For demodulation, the receiver must generate a carrier in phase and frequency synchronism to make a synchronous or coherent (also homodyne) demodulators.

Example 4.3 Switching Demodulator

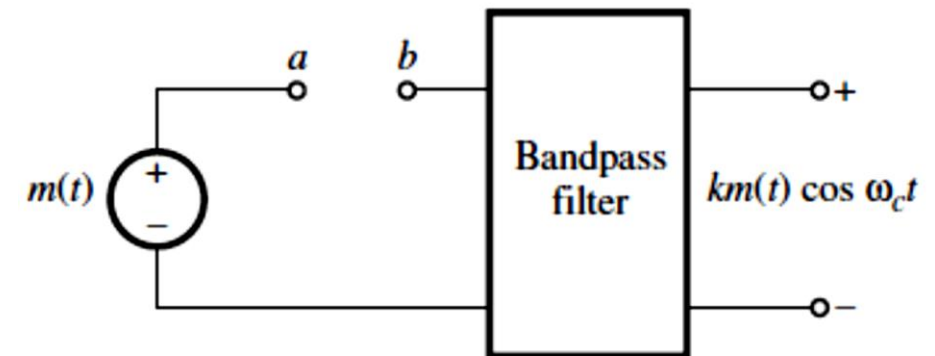
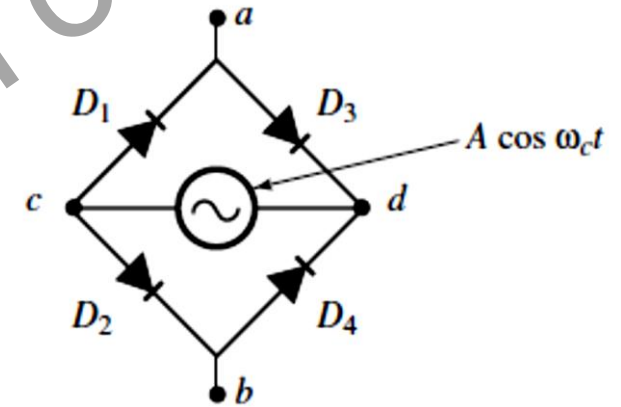
Analyze the switching demodulator that uses the electronic switch (diode bridge) in Fig. 4.5 as a switch (either in series or in parallel).

The input signal is $m(t)\cos\omega_c t$. The carrier causes the periodic switching on and off of the input signal. Therefore, the output is $m(t)\cos\omega_c t \times w(t)$. Using the identity $\cos x \cos y = 0.5[\cos(x+y) + \cos(x-y)]$, we obtain

$$\begin{aligned} m(t)\cos\omega_c t \times w(t) &= m(t)\cos\omega_c t \left[\frac{1}{2} + \frac{2}{\pi} \left(\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \dots \right) \right] \\ &= \frac{2}{\pi} m(t)\cos^2\omega_c t + \text{terms of the form } m(t)\cos n\omega_c t \\ &= \frac{1}{\pi} m(t) + \frac{1}{\pi} m(t)\cos 2\omega_c t + \text{terms of the form } m(t)\cos n\omega_c t \end{aligned}$$

- Spectra of the terms of the form $m(t)\cos n\omega_c t$ are centered at $\pm n\omega_c$ and are filtered out by the low pass filter LPF yielding the output $(1/\pi)m(t)$.

➤ Show that the output of a ring demodulator is $(2/\pi)m(t)$.



Amplitude Modulation with Carrier

- For the *suppressed carrier schemes*, the *major draw back* is that the receiver (Rx) has to generate a carrier in phase and frequency synchronism with the carrier at the transmitter (Tx) that may be located hundreds or thousands of miles away.
- The other alternative is to transmit a carrier $A \cos \omega_c t$ [along with the modulated signal $m(t) \cos \omega_c t$] so that there is no need to generate a carrier at the receiver.
- There is trade-off between complex receivers and high-power transmission in making a choice between point-to-point and broadcast communication systems.
- For broadcast communication, one high-power Tx and simpler, less expensive multiple Rx is the desirable trade-off that allows transmitting the carrier along with the modulated signal, known as conventional AM.

Amplitude Modulation with Carrier

➤ In time domain, the transmitted signal $\varphi_{AM}(t)$ is given by,

$$\varphi_{AM}(t) = A \cos \omega_c t + m(t) \cos \omega_c t$$

$$\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t$$

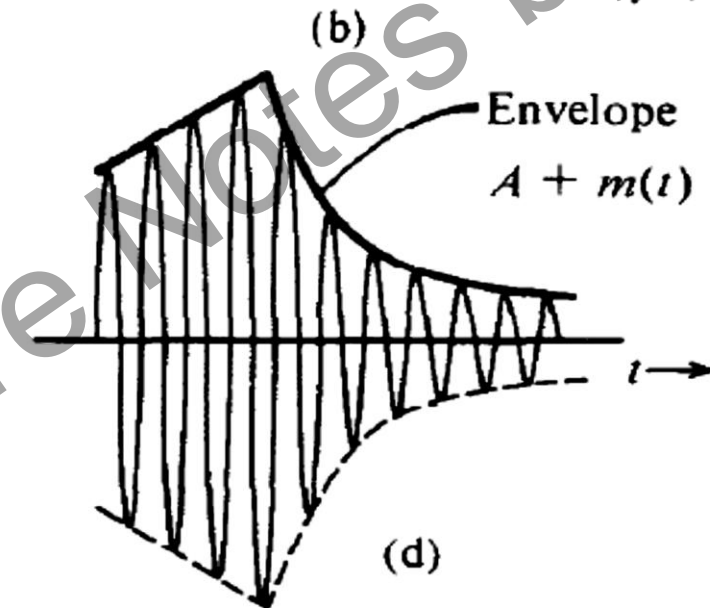
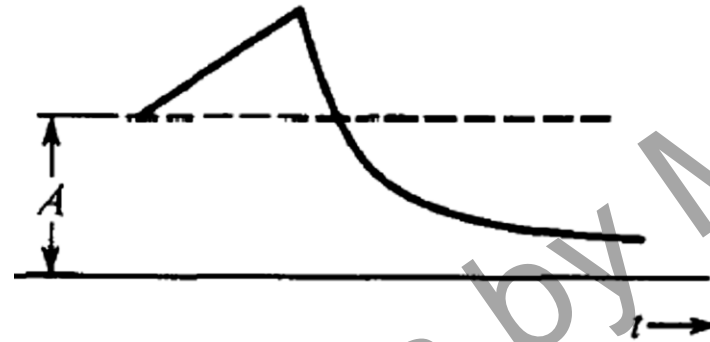
❖ In frequency domain, the spectrum of $\varphi_{AM}(t)$ is the same as that of $m(t) \cos \omega_c t$ plus two additional impulses at $\pm \omega_c$.

$$\begin{aligned} \varphi_{AM}(t) \\ \Leftrightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)] + \pi A [\delta(\omega + \omega_c) + \delta(\omega - \omega_c)] \end{aligned}$$

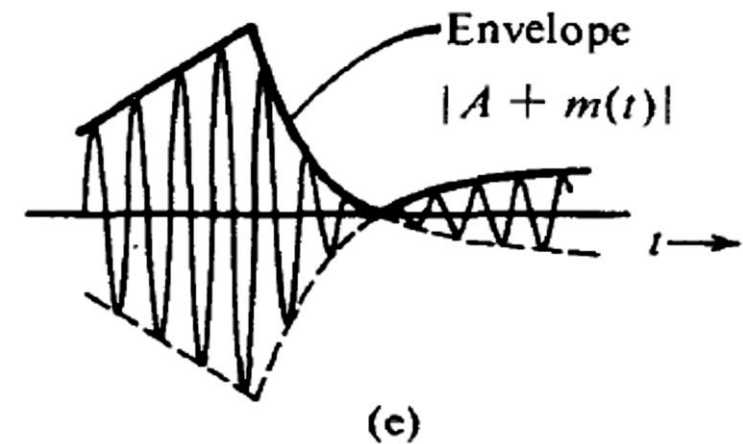
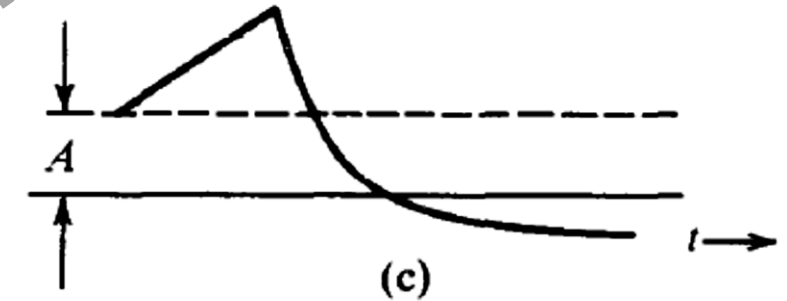
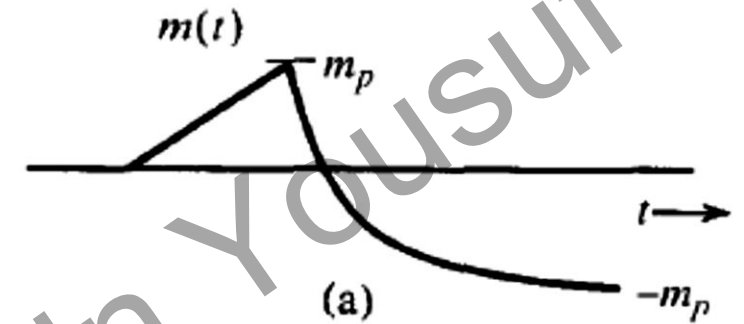
Recall that the AM signal is identical to the DSB-SC signal with $A + m(t)$ as the *modulating signal* [instead of $m(t)$].

Amplitude Modulation with Carrier

Case I: $A + m(t) > 0$ for all t

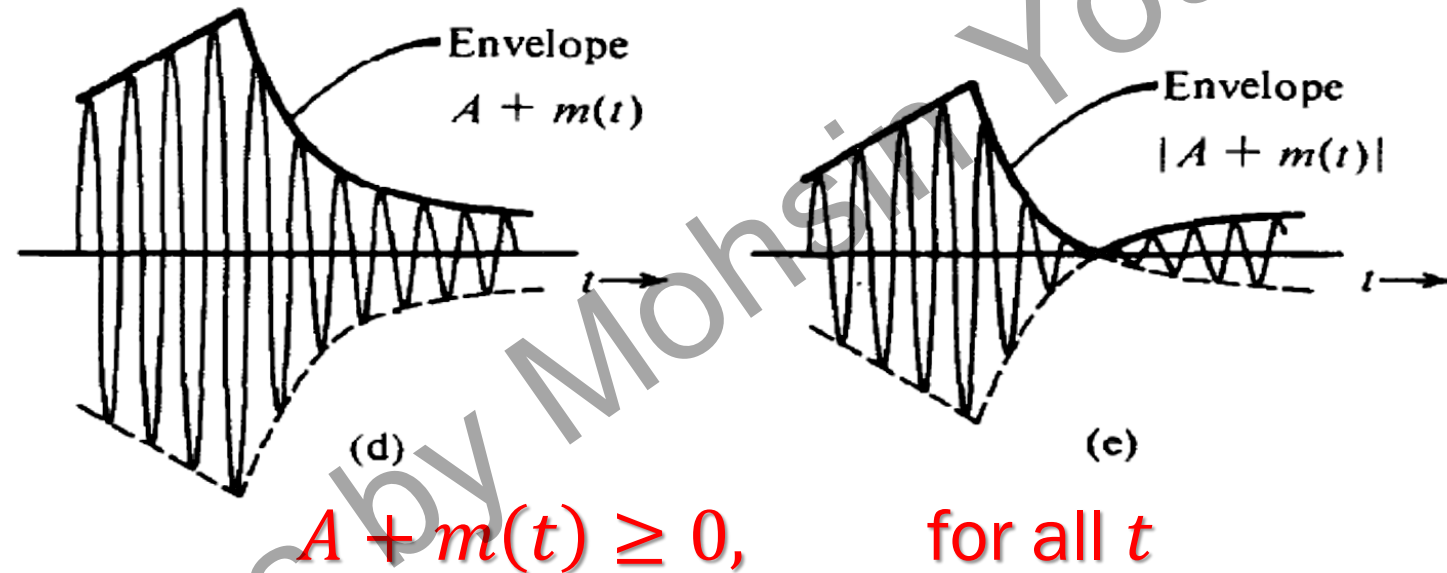


Case II: $A + m(t) \geq 0$ for all t



Amplitude Modulation with Carrier

Envelope Detection



CASE I: If $m(t) \geq 0$ for all t

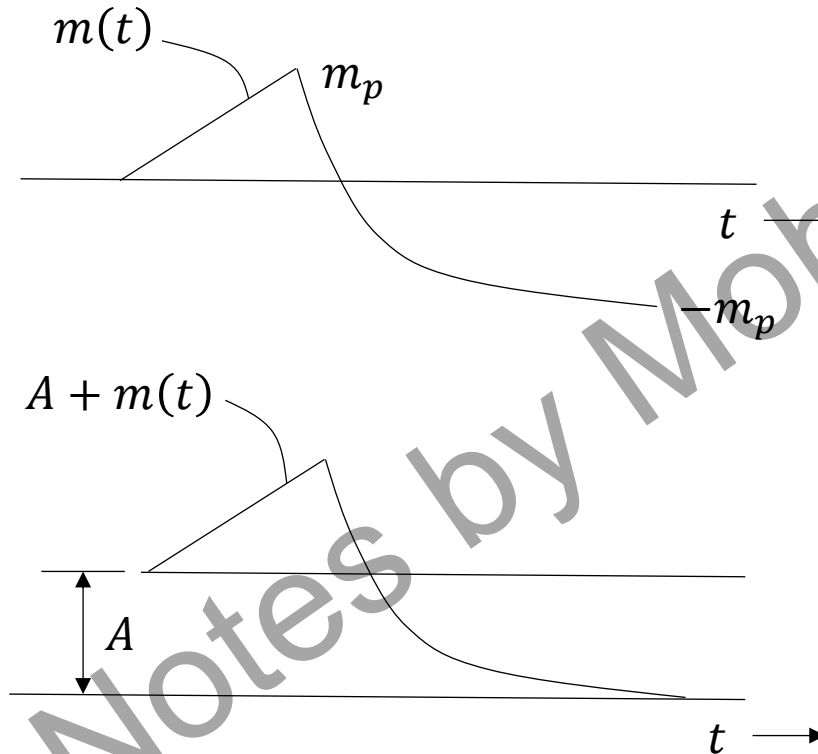
then $A = 0$ also satisfies the condition, no need to add any carrier because the envelope of the DSB-SC signal $m(t) \cos \omega_c t$ is $m(t)$ and such a signal can be detected by **envelope detection**.

CASE II: If $m(t) \not\geq 0$ for all t

Let m_p be the peak amplitude (+ive or -ive) of $m(t)$. This means that $m(t) \geq -m_p$. The above condition is $A \geq m_p$. Thus the minimum carrier amplitude for the viability of **envelope detection** is m_p .

Amplitude Modulation with Carrier

Envelope Detection



Condition for Envelope Detection

$$A + m(t) \geq 0 \text{ for all } t \quad (1)$$

$$-m_p \leq m(t) \leq m_p$$

$$A \geq -m_{\min} = m_p$$

$$A \geq m_p \quad (2)$$

Modulation Index

$$\mu = \frac{m_p}{A}$$

Range $0 \leq \mu \leq 1$

This is the required condition for the viability of demodulation of AM by an envelope detector.

Example 4.4 Tone Modulation and μ

Sketch $\varphi_{AM}(t)$ for modulation indices of $\mu = 0.5$ and $\mu = 1$, when $m(t) = B \cos \omega_m t$. This case is referred to as **tone modulation** because the modulating signal is a pure sinusoid (or tone).

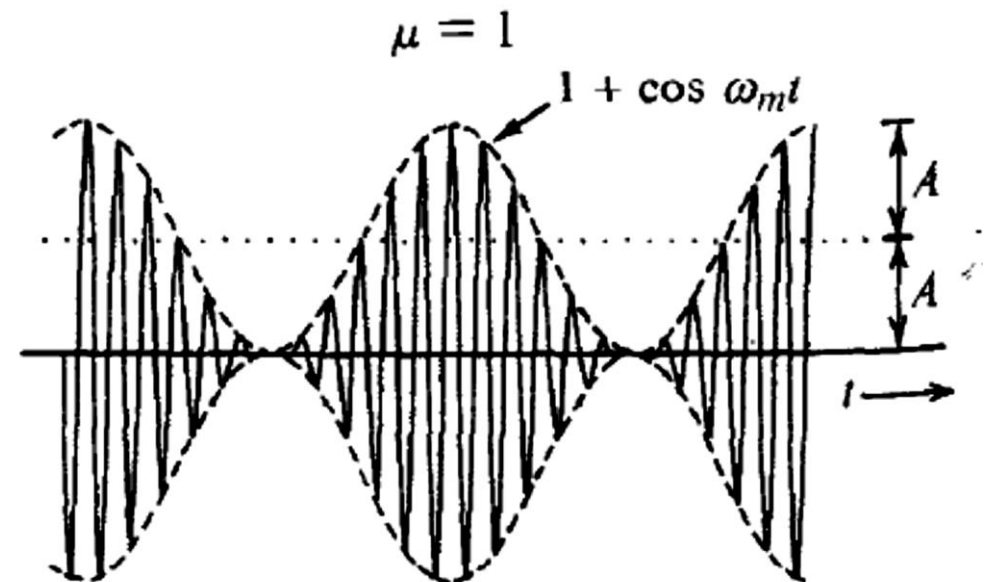
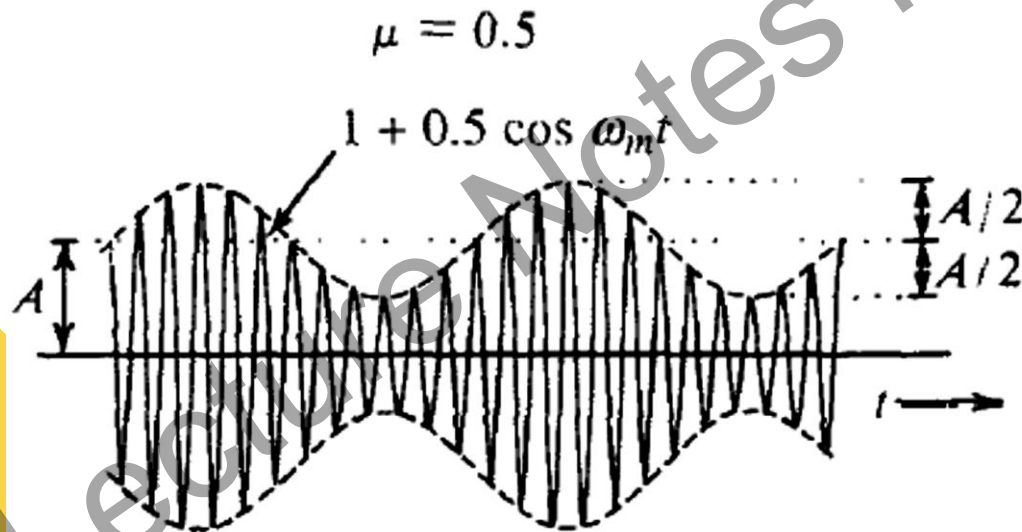
In this case, $m_p = B$ and the modulation index according to Eq. (4.10a) is

$$\mu = \frac{B}{A}$$

Hence, $B = \mu A$ and

$$m(t) = B \cos \omega_m t = \mu A \cos \omega_m t$$

$$\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t = A[1 + \mu \cos \omega_m t] \cos \omega_c t$$



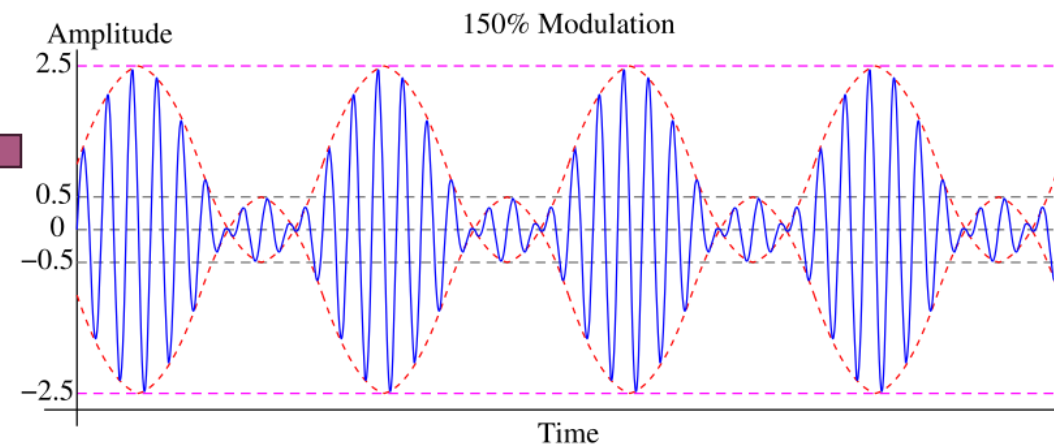
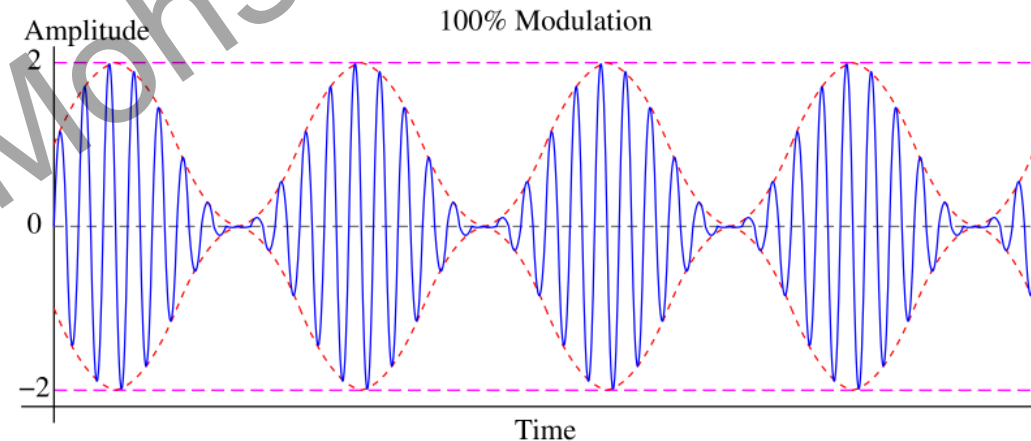
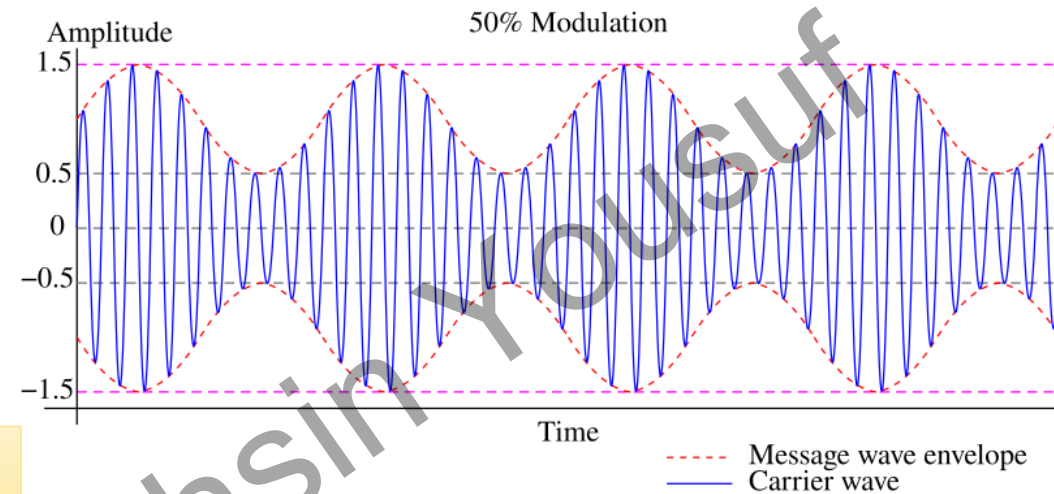
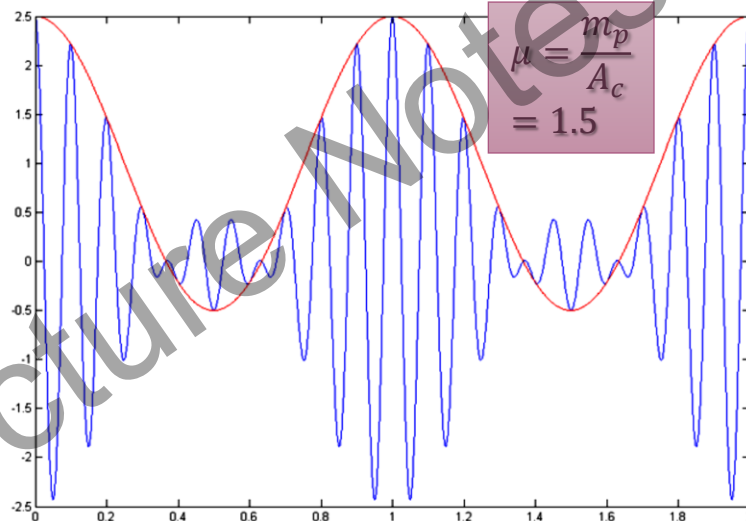
Amplitude Modulation and Modulation Index, μ

$$\varphi_{AM}(t) = A_c[1 + \mu \cos \omega_m t] \cos \omega_c t$$

$$\mu = \frac{m_p}{A_c}$$

Overmodulation
or
Envelope Distortion

$$\mu = \frac{m_p}{A_c} = 1.5$$



Amplitude Modulation with Carrier

The advantage of envelope detection in AM has its price. In AM, the carrier term does not carry any information, and hence, the carrier power is wasted,

$$\varphi_{AM}(t) = \underbrace{A \cos \omega_c t}_{\text{carrier}} + \underbrace{m(t) \cos \omega_c t}_{\text{sidebands}}$$

The power is calculated using mean square value of the signal, $P = \frac{1}{T} \int_0^T p \, dt$

The carrier power, $P_c = \frac{A^2}{2}$  Wasted power

The sideband power, $P_s = \frac{1}{2} \overline{m^2(t)}$  Useful power

The total power, $P_T = P_c + P_s$

The power efficiency, η is,

$$\eta = \frac{\text{useful power}}{\text{total power}} = \frac{P_s}{P_T} = \frac{P_s}{P_c + P_s} = \frac{\overline{m^2(t)}}{A^2 + \overline{m^2(t)}} 100\%$$

For the special case of tone modulation,

$$m(t) = \mu A \cos \omega_m t, \quad \text{and,} \quad \overline{m^2(t)} = \frac{(\mu A)^2}{2}$$

$$\eta = \frac{\mu^2}{2 + \mu^2} 100\%, \quad \text{and,} \quad \eta_{\max} = 33\%$$

Example 4.5 Tone Modulation and Power

Determine η and the percentage of the total power carried by the sidebands of the AM wave for tone modulation when (a) $\mu = 0.5$ and (b) $\mu = 0.3$.

For $\mu = 0.5$,

$$\eta = \frac{\mu^2}{2 + \mu^2} 100\% = \frac{(0.5)^2}{2 + (0.5)^2} 100\% = 11.11\%$$

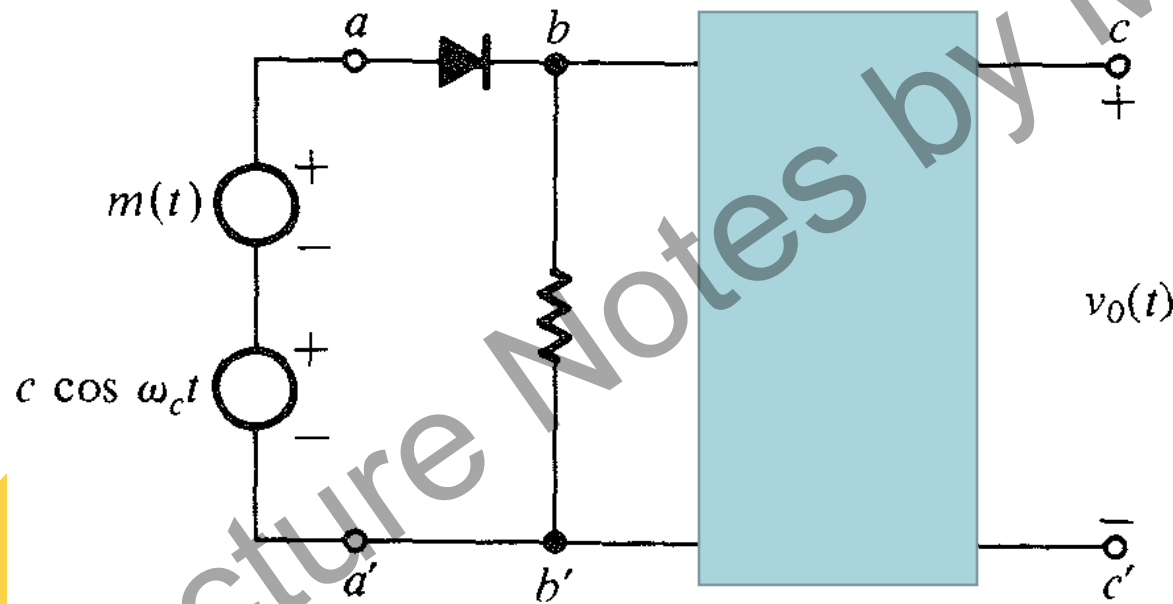
Hence, only about 11% of the total power is in the sidebands. For $\mu = 0.3$,

$$\eta = \frac{(0.3)^2}{2 + (0.3)^2} 100\% = 4.3\%$$

Hence, only 4.3% of the total power is the useful power (power in sidebands).

Generation of Amplitude Modulation with Carrier

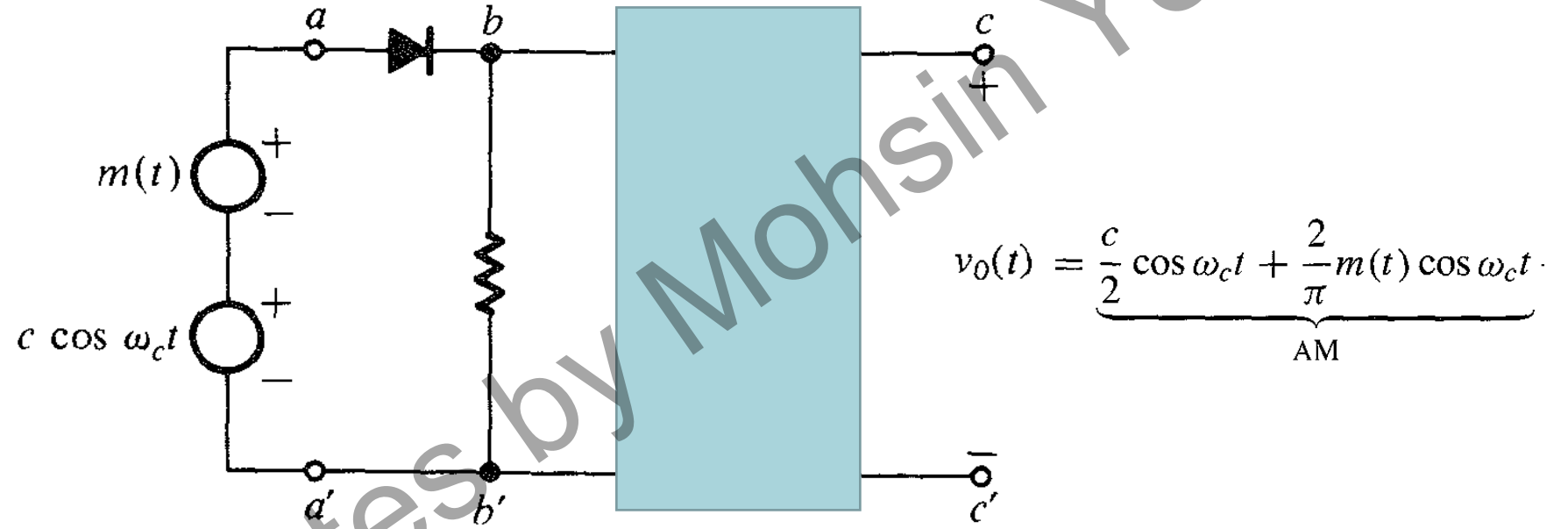
- AM Signals can be generated using any of the DSB-SC modulators if the modulating signal is $A + m(t)$ instead of just $m(t)$.
- No need to have a balanced circuit, accounts to a simple modulators for AM.



Specifications:

1. Single diode instead of a bridge
2. Input is $[c \cos \omega_c t + m(t)]$
3. Now, $c \gg m(t)$, Why?
4. Switching action of the diode is controlled by $c \cos \omega_c t$

Generation of Amplitude Modulation with Carrier



$$v_0(t) = \underbrace{\frac{c}{2} \cos \omega_c t + \frac{2}{\pi} m(t) \cos \omega_c t}_{\text{AM}}$$

$$v_{bb'}(t) = [c \cos \omega_c t + m(t)]w(t)$$

$$= [c \cos \omega_c t + m(t)] \left[\frac{1}{2} + \frac{2}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right) \right]$$

$$= \underbrace{\frac{c}{2} \cos \omega_c t + \frac{2}{\pi} m(t) \cos \omega_c t}_{\text{AM}} + \underbrace{\text{other terms}}_{\text{blue box}}$$

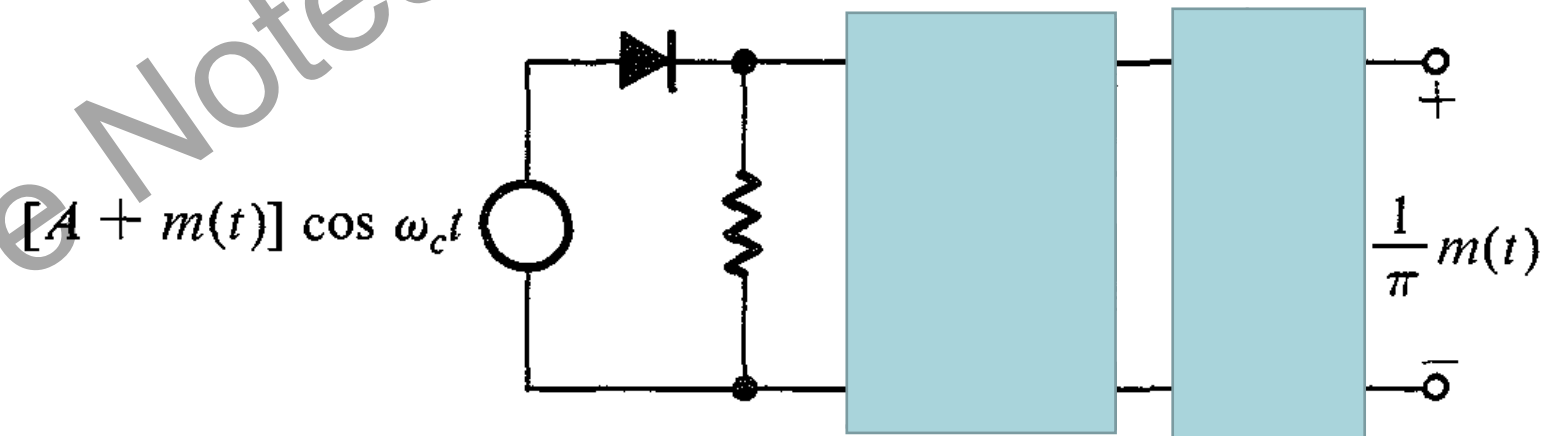
Demodulation of Amplitude Modulation with Carrier

- AM Signals can be demodulated **coherently** by using a locally generated carrier but will defeat the very purpose of AM. Why?

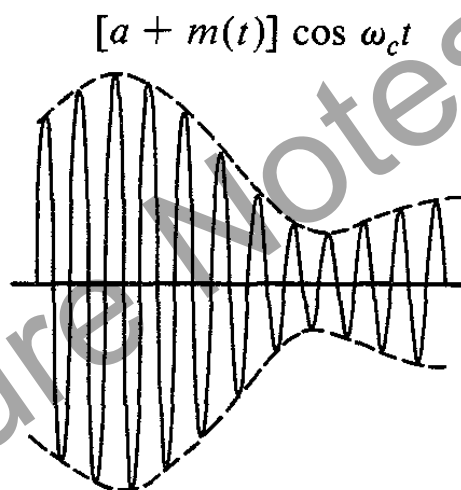
Non-coherent methods of AM demodulation

1. Rectifier Detection

2. Envelope Detection



1. Rectifier Detection



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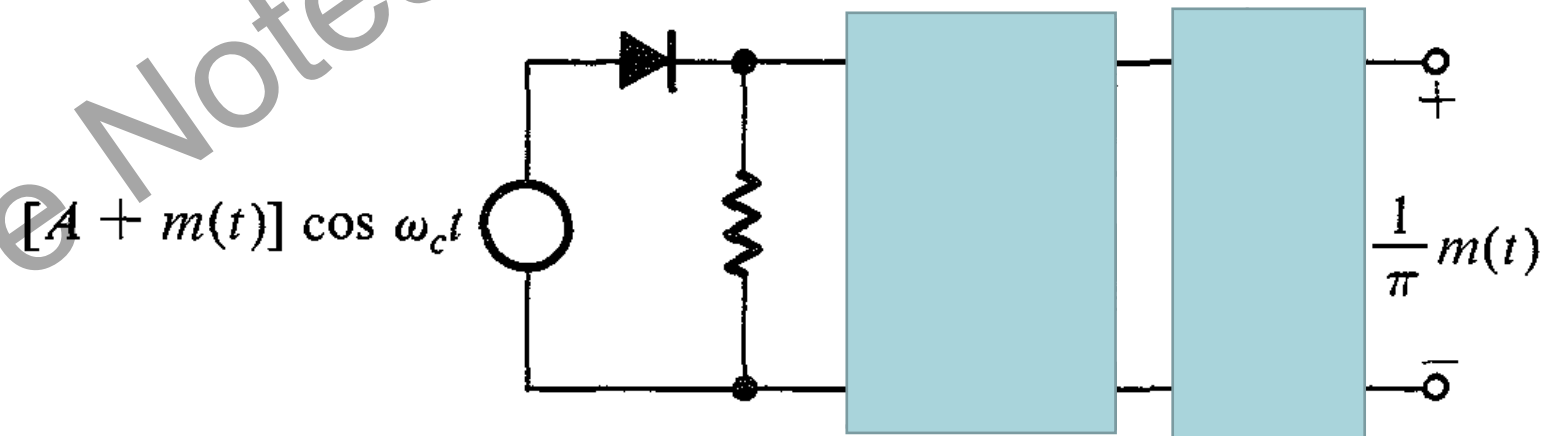
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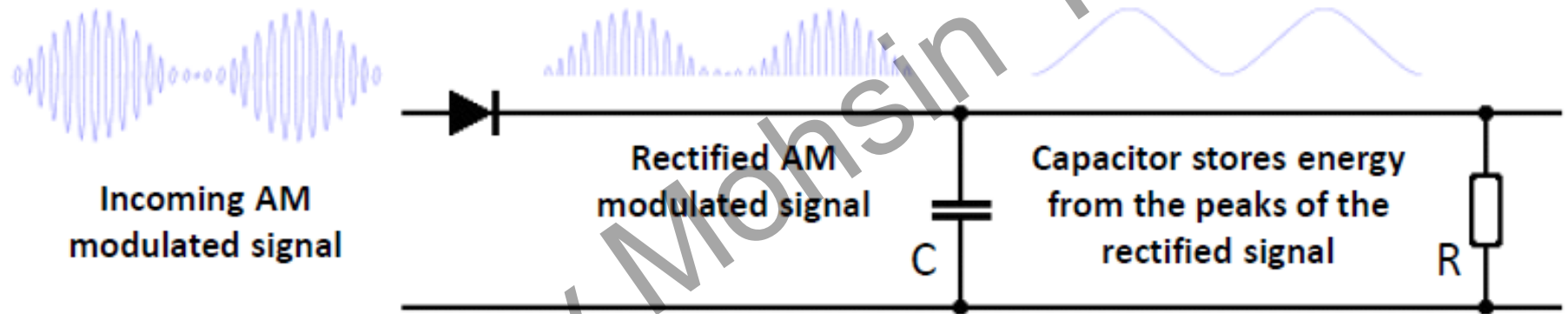
Non-coherent methods of AM demodulation

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Demodulation of Amplitude Modulation with Carrier



Key idea: Capacitor captures the voltage peaks of rectified waveform

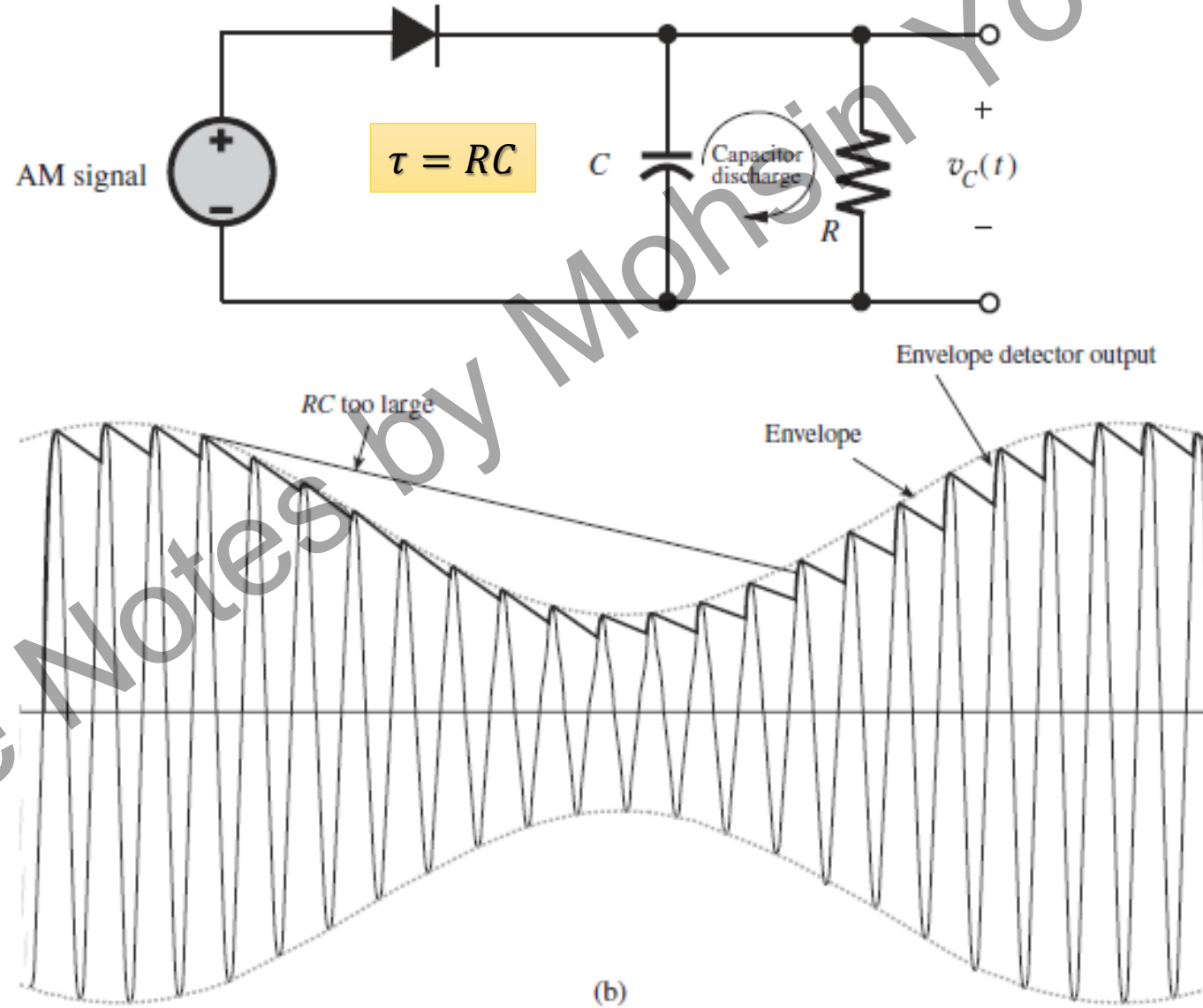
Envelope Detection requires the an RC network with time constant $\tau = RC$

Two conditions must be met for an envelope detector to work:

- (1) Narrowband [meaning $f_c \gg \text{bandwidth of } m(t)$]
- (2) $A_C + m(t) \geq 0$

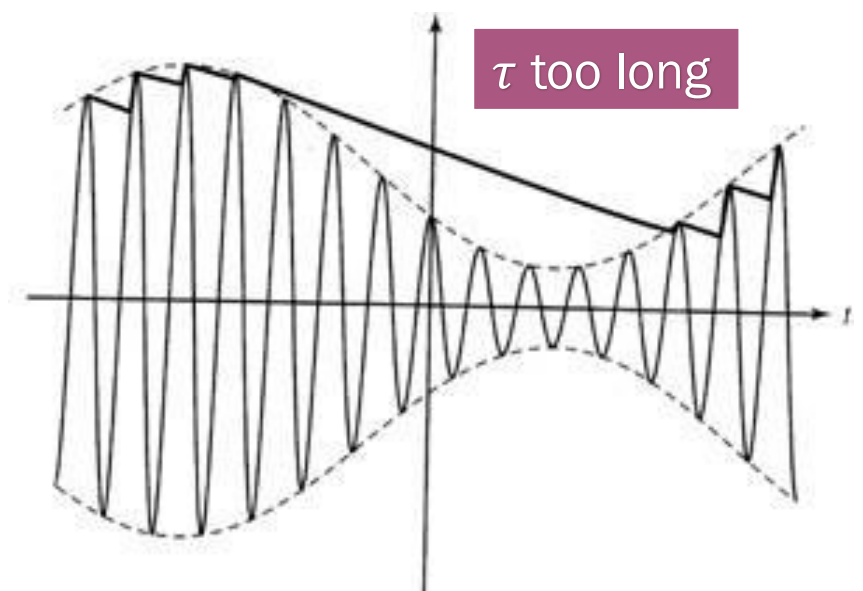
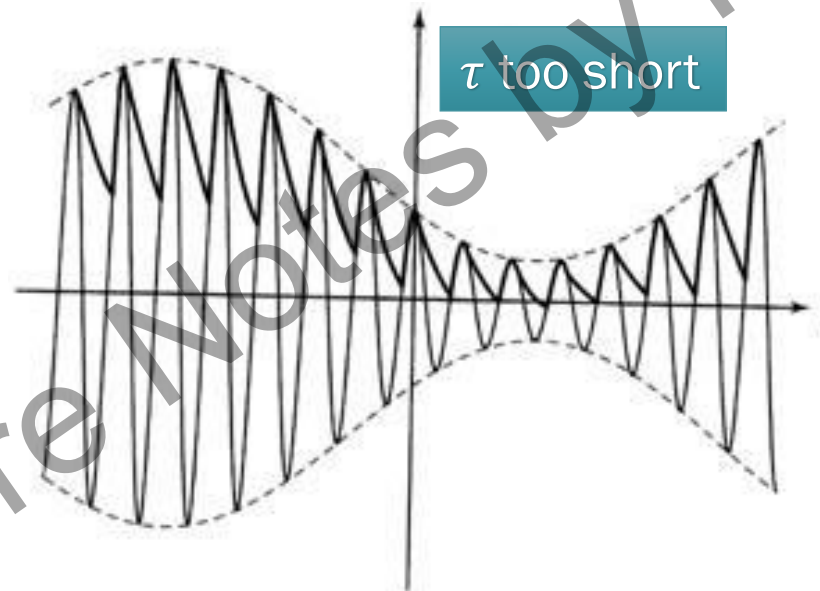
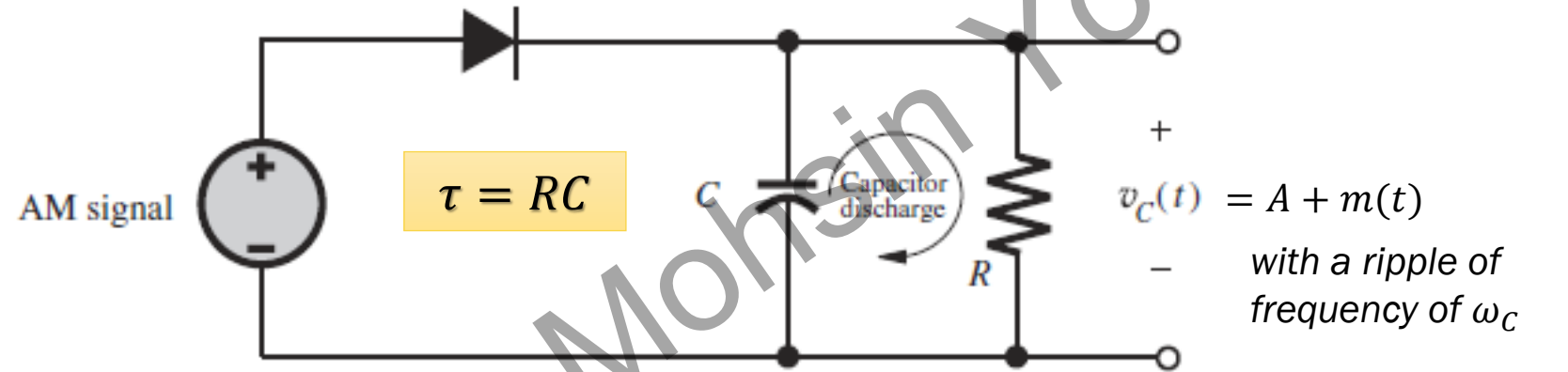
Demodulation of Amplitude Modulation with Carrier

2. Envelope Detection



Demodulation of Amplitude Modulation with Carrier

2. Envelope Detection



Design Criterion for RC

$$1/\omega_C \ll RC < 1/(2\pi B)$$

$$2\pi B < \frac{1}{RC} \ll \omega_C$$

Demodulation of Amplitude Modulation with Carrier

- Both the **rectifier detector** and the **envelope detector** consist of a **half-wave rectifier** followed by a **low-pass filter (LPF)**.
- Superficially, these detectors may appear equivalent, but they are distinct and operate on very different principles.

Non-coherent methods of AM demodulation

1. Rectifier Detection

Synchronous Demodulator
(Does not depend on μ)

2. Envelope Detection

Nonlinear Operation
(Does depend on μ)

Example 4.6 Tone Modulation for Envelope Detection

For tone modulation (Example 4.4), determine the upper limit of RC to ensure that the capacitor voltage follows the envelope.

$$v_C = E e^{-t/RC}$$

Because the time constant is much larger than the interval between the two successive cycles of the carrier ($RC \gg 1/\omega_c$), the capacitor voltage v_C discharges exponentially for a short time compared to its time constant. Hence, the exponential can be approximated by a straight line obtained from the first two terms in Taylor's series for $E e^{-t/RC}$,

$$v_C \simeq E \left(1 - \frac{t}{RC} \right)$$

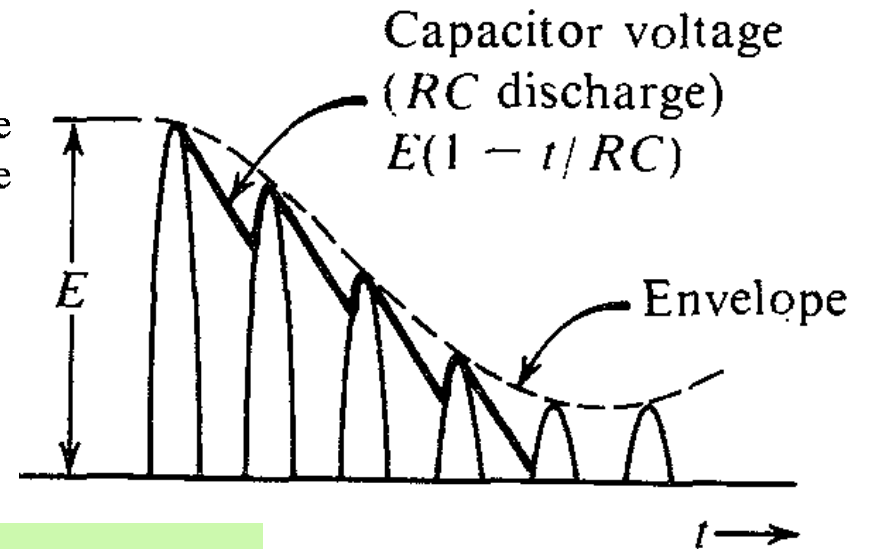
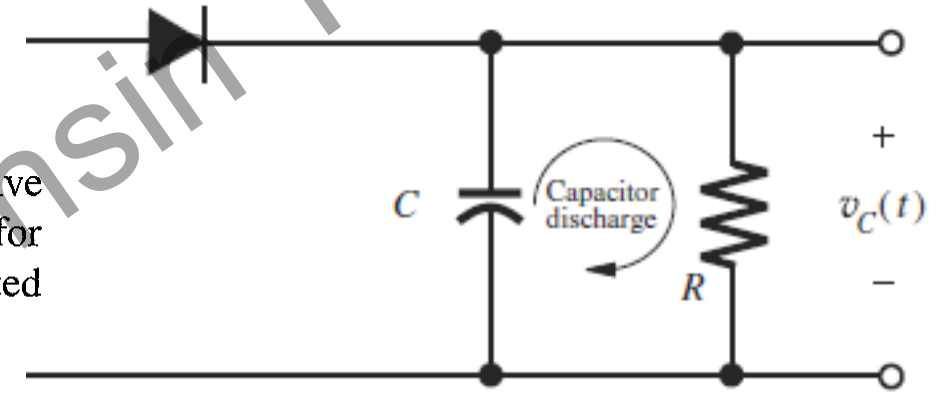
The slope of the discharge is $-E/RC$. In order for the capacitor to follow the envelope $E(t)$, the magnitude of the slope of the RC discharge must be greater than the magnitude of the slope of the envelope $E(t)$. Hence,

$$\left| \frac{dv_C}{dt} \right| = \frac{E}{RC} \geq \left| \frac{dE}{dt} \right|$$

But the envelope $E(t)$ of a tone-modulated carrier is [Eq. (4.11)]

$$E(t) = A[1 + \mu \cos \omega_m t]$$

$$\frac{dE}{dt} = -\mu A \omega_m \sin \omega_m t$$



Example 4.6 Tone Modulation for Envelope Detection

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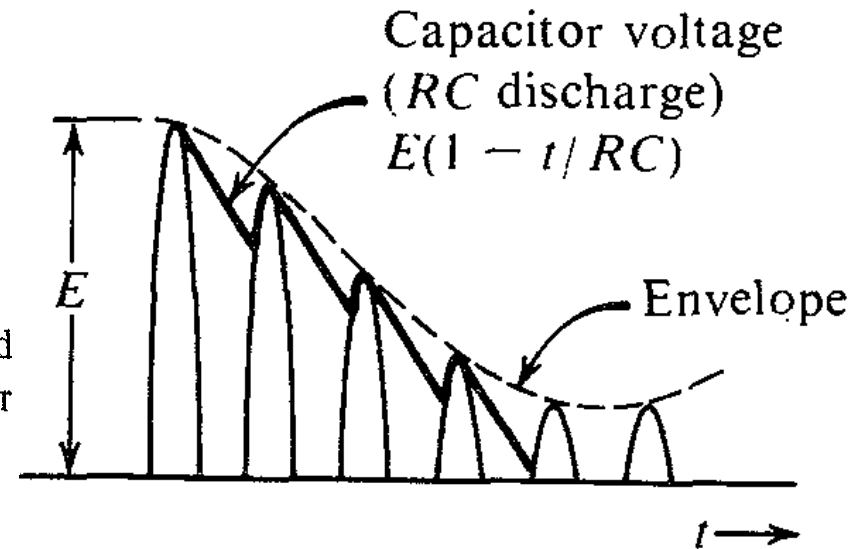
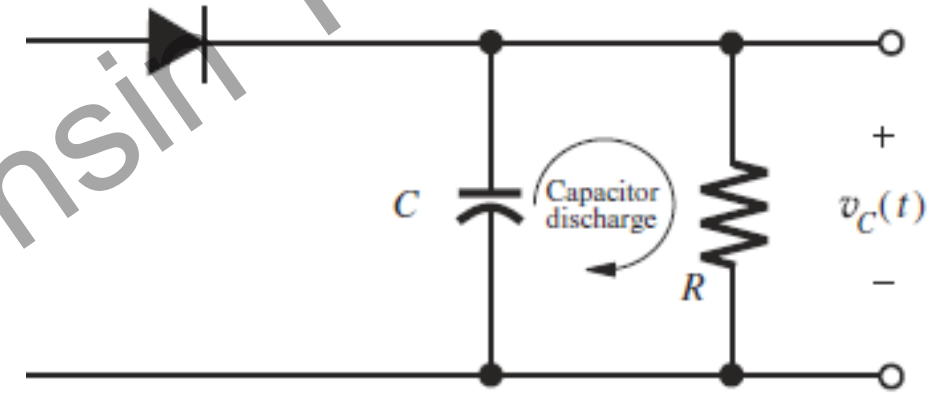
$$\left| \frac{dv_C}{dt} \right| = \frac{E}{RC} \geq \left| \frac{dE}{dt} \right|$$

$$\frac{A(1 + \mu \cos \omega_m t)}{RC} \geq \mu A \omega_m \sin \omega_m t \quad \text{for all } t$$

$$RC \leq \frac{1 + \mu \cos \omega_m t}{\mu \omega_m \sin \omega_m t} \quad \text{for all } t$$

The worst possible case occurs when the right-hand side is the minimum. This is found (as usual, by taking the derivative and setting it to zero) to be when $\cos \omega_m t = -\mu$. For this case, the right-hand side is $\sqrt{(1 - \mu^2)}/\mu \omega_m$. Hence,

$$RC \leq \frac{1}{\omega_m} \left(\frac{\sqrt{1 - \mu^2}}{\mu} \right)$$



Preview of AM Categories

